

Hua

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Education

University of Illinois at Urbana-Champaign | Champaign, IL

Expected May 2025

Bachelor of Science in Statistics & Kinesiology (Dual Degree)

Cumulative GPA: 3.79/4.00

Courses: Statistical Analysis, Statistics and Probability, Statistics Programming Methods, Linear Algebra with Computer Application, Statistical Data Management, Statistical Modeling, Unsupervised Machine Learning, Advanced Data Analysis

Skill

Programming and Software: Python, SQL, R, Matlab, Tableau, SAS, MS Office Suite, Google Suite, Bash

Machine Learning Techniques: Decision Tree, Random Forest, Bagging, Boosting, clustering, PCA (Principal Component Analysis), Regularization, Model Evaluation, Exploratory Data Analysis, Experimental Design, Hypothesis Testing

Internship Experience

ATLAS | Data Analyst | Champaign, IL

Aug - Dec 2023

- Identified key email topics, such as UI registration and server operating system-related issues, conducted keyword analysis to reveal usage patterns, and observed enhanced communication
- Preprocessed email subject data in **Excel**, used **Tableau** to process visualization and extract the top 5 topics
- Presented analysis results and key findings through slides to 10 technical and non-technical stakeholders
- Generated a database with Python to record and extract historical data, improving operating efficiency by 15%

Raybiotech Life, Inc. | Research Assistant | Guangzhou, China

Jun - Jul 2021

- Fused sp2/0 (myeloma cells) with B lymphocytes, creating hybridomas capable of unlimited growth
- Filtrated, expanded, and scaled up antibody cells from 96-well microtiter plates to individual petri dishes based on designed procedures
- Completed the entire process of antibody production, from resuscitating myeloma cells in liquid nitrogen to injecting antibody cells into mice and collecting ascites

Project Experience

Credit Card Fraud Prediction (Python)

Aug 2024

- Spearheaded a credit card fraud detection project, leveraging Random Forest, XGBoost, LightGBM, and CatBoost algorithms, to achieve superior model performance over traditional regression methods
- Conducted a rigorous model selection process by comparing multiple algorithms using confusion matrices, ROC & AUC and Precision-Recall Curves, optimizing XGBoost model for the best performance
- Achieved a model recall of 83.59%, successfully identifying approximately 83.6% of all fraudulent transactions
- Delivered significant business impact by reducing potential fraud losses by an estimated \$8.36 million annually

Predictive Modeling for Student Weekend Alcohol Consumption (Python)

Nov - Dec 2023

- Developed and implemented a binary classification model to predict student weekend alcohol consumption levels
- Processed data and optimized model with k-Fold cross-validation for accurate predictions
- Achieved a notable test AUC of 0.8403 with the developed model, identifying students at risk of problematic drinking

Interactive 5-Day Weather Forecast Dashboard (R, API interactions, HTTP requests, JSON parsing)

Nov - Dec 2023

- Built a Shiny-based 5-Day Weather Forecast Dashboard, integrating real-time weather data from the OpenWeatherMap API with secure API key authentication
- Developed a user-friendly interface, featuring interactive plot with **ggplot2** and action button for controlled data retrieval
- Employed reactive programming principles to ensure a responsive user interface, updating results only upon user-triggered actions and enabling the selection of temperature scales (Celsius or Fahrenheit)
- Engineered a summary table utilizing **tidyverse** for data manipulation to present key weather statistics, such as average temperature, feels-like temperature and extremes, organized by date

Starbucks Customer Data Analytics (SQL)

Jul - Aug 2023

- Conversed and structured Starbucks' raw customer data, and used **SQL** queries to analyze customer behavior trends, revealing correlations between demographic attributes and offer completion rates for targeted marketing strategies
- Leveraged advanced **JSON** to extract and process data within the database to streamline information retrieval and analysis, and employed **window functions** for enhanced query efficiency and optimized data processing
- Assessed customer engagement by segmenting demographics, identifying key segments that exhibited higher offer completion rates to optimize engagement strategies

Research Experience

AI Farm: Automated Phenotype Monitoring (From Plots to Fields) | Champaign, IL

May - Dec 2023

- Utilized AI to enhance crop genetics and optimize agricultural practices, employing drone imaging and data for field-level insights
- Devised a comprehensive strategy to detect upward-facing leaves for accurate leaf width measurements and utilized VGG Image Annotator to label leaves to train a neural network, paving the way for robust leaf identification
- Trained the Mask R-CNN model using labeled images to improve model accuracy and performance
- Labeled 100+ leaves and bolstered the model's training dataset to include 700+ leaves, achieving Intersection over Union of about 0.7 for heightened accuracy in object detection and segmentation